DSC 40B - Homework 05

Due: Tuesday, November 3

Instructions:

- Write your solutions to the following problems **by hand**, either on paper that you scan or using a tablet. Typed solutions will not be accepted for credit!
  - Code is an exception. You do not need to handwrite code; you can instead include it (and any plots it makes) as a screenshot.
- Unless otherwise noted by the problem's instructions, show your work or provide some justification for your answer.
- Homework problems are graded pass/fail on effort, not correctness.
- Homeworks are due via Gradescope at 11:59 PM.

This homework is shorter – just one question – due to the midterm exam. It's also worth fewer credits than usual.

Programming Problem 1. (1 credit)

Recall that a *mode* of a collection is an element which occurs with the greatest frequency. For example, 4 is a mode of the collection $4, 5, 8, 3, 4, 2, 4, 5, 5, -2$. 5 is also a mode, since it occurs just as frequently as 4.

In a file named `mode.py`, create a function named `mode(numbers)` which takes in an array of numbers and returns a mode. Your function should have the best possible expected time complexity. If there are multiple modes, your function need only return one of them. You should *not* assume that the numbers are integers or that they are positive.

Note: it's up to you to determine the best possible time complexity and convince yourself that your code is efficient! Think back to when we talked about theoretical lower bounds. Can you come up with one for this problem? This is a situation you'll find yourself in often when writing code: there's usually no resource that tells you what the best time complexity is for a given problem, so you'll have to convince yourself that your solution is optimal.

As this is a programming problem, submit your `mode.py` file to the Gradescope autograder assignment named "Homework 05 - Programming Problem 01". Since this is the homework's only problem, there is no written part to submit.